

ClearEval: A Database-Grounded Feasibility Benchmark for LLM-Generated Tissue Optical Clearing Protocols

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Abstract—Tissue Optical Clearing (TOC) enables non-destructive, high-resolution 3D imaging, but protocol design requires compatible choices of clearing method, labeling strategy, refractive-index condition, and incubation time. While LLMs perform well on biomedical knowledge tasks, their ability to generate feasible wet-lab protocols remains under-evaluated. We introduce ClearEval, a database-grounded feasibility benchmark for LLM-generated TOC protocols. ClearEval is built on a structured TOC feasibility database integrating 372 wet-lab experimental records, 400+ curated tissue-clearing literature chunks, and expert-reviewed constraints on methods, tissues, markers, timing, and compatibility. It evaluates 13 state-of-the-art LLMs through two tracks: 2,588 multiple-choice questions for static knowledge and 253 open-ended scenarios for protocol generation. Generated protocols are scored by our Completeness, Correctness, and Effectiveness (CCE) framework, which combines rubric-based procedural assessment with database-grounded feasibility rules and formulas. Models reach 79–91% executable protocol generation; the strongest model falls from 91.1% Index to a 55.2% Application Index. The main failures arise from executable parameterization, especially matching fluorescent markers to required labeling targets. ClearEval provides a depth-first feasibility benchmark that complements breadth-oriented protocol benchmarks and supports pre-experimental evaluation in TOC and related wet-lab subdomains. Code and data: <https://anonymous.4open.science/r/ClearEval>.

Index Terms—Benchmark, Tissue optical clearing, Wet-lab protocols, Large language models

I. INTRODUCTION

Tissue Optical Clearing (TOC) renders biological tissues optically transparent by reducing light scattering and refractive-index mismatch, enabling non-destructive, high-resolution 3D imaging via light-sheet microscopy [1], [2]. However, designing a TOC protocol is not a single-answer task. Given a specimen, labeling target, imaging goal, and laboratory constraint, experimenters must choose a feasible combination of clearing method, labeling strategy, refractive-index condition, incubation schedule, sample geometry, and operational constraints. Multiple combinations may be acceptable, but one incompatibility—for example, a quenched fluorophore, an unsuitable method for the sample scale, or an unrealistic incubation time—can invalidate the protocol and irreversibly damage the specimen.

Existing biomedical LLM benchmarks, including MedQA [3], MMLU [4], and recent biomedical extensions [5]–[7], mainly evaluate knowledge retrieval,

question answering, or text-based reasoning. These settings measure whether models know biomedical facts, but they do not directly test whether a generated wet-lab protocol is feasible under experimental constraints. Likewise, semantic metrics such as BLEU [8] and general-purpose LLM evaluators [9], [10] do not explicitly evaluate method–label compatibility, refractive-index suitability, or timing plausibility. This creates a gap between knowledge evaluation and pre-experimental feasibility checking.

We introduce **ClearEval**, a database-grounded feasibility benchmark for LLM-generated TOC protocols. ClearEval integrates 372 wet-lab experimental records, 400+ curated tissue-clearing literature chunks, and expert-reviewed TOC constraints into a structured feasibility database. This database supports both task construction and protocol scoring. The benchmark contains 2,588 multiple-choice questions for static knowledge and 253 open-ended scenarios for protocol generation. Generated protocols are evaluated by our **CCE (Completeness, Correctness, Effectiveness)** framework, which combines rubric-based procedural assessment with database-grounded rules and formulas for experimental effectiveness.

Experiments on 13 state-of-the-art LLMs show that static knowledge performance is close to saturation, whereas open-ended feasibility evaluation remains discriminative. For example, GPT-5.2-Fast achieves 91.1% Index but only 55.2% on the Application Index (Table II); MCQ profiles are tightly clustered, while OEQ-based CCE scores expose model-specific execution failures. The most reliable failures arise from executable parameterization, especially matching fluorescent markers to the required labeling targets. Our contributions are: (1) a structured TOC feasibility database integrating experimental records, literature evidence, and expert constraints; (2) a dual-track benchmark and database-grounded CCE evaluator for LLM-generated TOC protocols; and (3) an empirical analysis showing that feasibility evaluation reveals execution-level failures hidden by static knowledge tasks.

II. RELATED WORKS

Scientific and biomedical LLM evaluation. The evaluation of scientific and biomedical LLMs has moved from general knowledge tests toward more specialized biomedical and research-oriented tasks. Benchmarks such as MMLU [4], MedQA [3], Med-PaLM-style evaluations [11], and recent

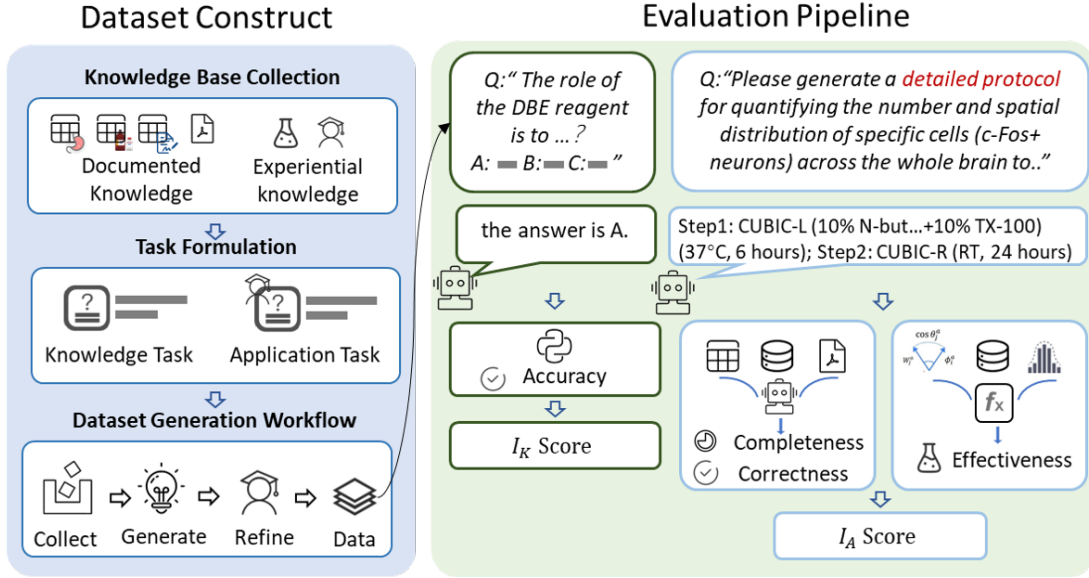


Fig. 1. Overview of ClearEval. A TOC feasibility database is constructed from wet-lab experimental records, curated literature chunks, and expert-reviewed constraints. The database supports both task construction and protocol scoring. The Knowledge track is scored by the Knowledge Index (I_K), while the Application track is scored by the Application Index (I_A), which combines Completeness, Correctness, and Effectiveness.

biomedical benchmarks [5]–[7] measure knowledge retrieval, clinical reasoning, biomedical NLP, and interactive question answering. These resources provide standardized evidence for biomedical competence, but their outputs are usually closed-form answers, short explanations, or task-specific labels rather than executable wet-lab protocols.

Biological protocol and wet-lab benchmarks. Recent benchmarks have extended LLM evaluation toward biological protocols and laboratory workflows. BioProBench [12] studies biological protocol understanding and reasoning across multiple subfields, including protocol QA, step ordering, error correction, generation, and reasoning. LAB-Bench [13] evaluates language models on biology research tasks, including literature use, protocol planning, and data analysis. BixBench [14] considers multi-step computational-biology agent tasks. SeedBench [15] similarly targets a single wet-lab subdomain (seed science) but evaluates multi-task knowledge rather than parameter-level protocol feasibility. These benchmarks characterize broad biological or research capabilities. ClearEval focuses on a complementary setting: parameter-level feasibility assessment for LLM-generated Tissue Optical Clearing protocols, where multiple method–marker–time combinations may be acceptable and must be checked against explicit experimental constraints.

Automatic evaluation, TOC, and pre-experimental checking. Automatic evaluation methods, including text-overlap metrics such as BLEU [8] and LLM-based evaluators such as G-Eval [9], enable scalable assessment of generated outputs. Recent work has also analyzed systematic effects in LLM-as-a-judge methods [10], [16]. In ClearEval, LLM judgment is used for rubric-defined procedural dimensions, while Effectiveness is computed from a structured TOC feasibility database. TOC

methods (CLARITY, CUBIC, iDISCO, uDISCO, PEGASOS) enable high-resolution 3D imaging across tissue and organ scales [1], [2], [17]–[19]; ClearEval organizes this literature into a machine-readable feasibility checker for LLM-generated protocols, a pre-experimental component for vertical scientific agents [20], [21].

III. THE CLEAREVAL BENCHMARK

ClearEval is organized around three linked components: a TOC feasibility database, a dual-track task suite, and a scoring framework (Fig. 1). The feasibility database converts experimental records, literature evidence, and expert-reviewed TOC constraints into machine-readable resources. The task suite uses these resources to construct knowledge questions and open-ended protocol-design scenarios. The scoring framework evaluates model outputs with a Knowledge Index (I_K) for MCQs and an Application Index (I_A) for generated protocols.

A. TOC Feasibility Database

ClearEval formalizes TOC protocol constraints as a structured feasibility database. The database integrates 372 wet-lab experimental records used to support clearing-time intervals, 400+ curated tissue-clearing literature chunks, and expert-reviewed constraints on tissues, molecular targets, fluorescent markers, clearing-method properties, refractive-index conditions, sample scales, and compatibility rules. These resources are encoded into seven machine-readable knowledge bases, with inventory, provenance, and review status documented in Appendix A.

The database is designed to support feasibility assessment rather than single-answer protocol matching. For method selection, each clearing method is represented along six experi-

mental axes: fluorescence/protein preservation, dye permeability, clearing capacity, sample geometry, operational economy, and biosafety. These axes provide a shared constraint space for comparing the requirements of an experimental scenario with the properties of candidate clearing strategies.

B. Task Suite Construction

ClearEval contains two task tracks. The Knowledge track evaluates static TOC knowledge. Expert-authored templates are instantiated against the feasibility database through controlled-vocabulary substitution, producing 2,588 multiple-choice questions: 533 named-entity questions, 1,350 domain-knowledge questions, and 705 task-scenario questions. Each question uses an 8-option format, followed by deduplication and option shuffling.

The Application track evaluates open-ended TOC protocol generation. We define 60 expert-designed scientific missions across biological scales and combine them with the feasibility database to construct constrained experimental scenarios. These scenarios are further augmented with real wet-lab challenges and quality-filtered to 253 open-ended tasks. Because TOC protocol design may admit multiple feasible method–marker–time combinations, these tasks are not graded by reference-text similarity. They are instead evaluated by the scoring framework below.

C. Scoring Framework

ClearEval scores model outputs at two levels (Table I). The Knowledge Index (I_K) is the macro-averaged accuracy over MCQ categories. The Application Index (I_A) evaluates open-ended generated protocols through three dimensions: Completeness, Correctness, and Effectiveness:

$$I_A = \min(\text{Completeness}, \text{Correctness}, \text{Effectiveness}). \quad (1)$$

This bottleneck formulation reflects wet-lab practice: a protocol can be detailed and fluent, yet still fail if one essential component is missing, inconsistent, or experimentally infeasible. Thus, CCE denotes the scoring structure of the Application track, whereas ClearEval as a whole combines both static knowledge evaluation and application-level feasibility assessment.

a) Completeness and Correctness.: Completeness measures whether a protocol specifies a runnable TOC experiment, via Step Integrity (are essential stages—fixation/pre-treatment, labeling, clearing, RI matching, imaging preparation—present when applicable) and Parameter Detail (are key parameters given: reagent/method choice, sample scale, incubation time, temperature, concentration, labeling, imaging settings). Correctness measures internal consistency with TOC principles through four equally-weighted sub-checks: Sequential Logic (co_order , plausible wet-lab ordering); Method Consistency (co_method , method matches goal and sample); Parameter Consistency (co_param , scale/timing/concentration/temperature mutually consistent and respecting diffusion limits); and Chemical Safety (co_chem , no introduced non-standard hazard). Both

TABLE I
THE HIERARCHICAL METRIC SYSTEM OF CLEARVAL. EVALUATION SCALES FROM STATIC KNOWLEDGE RETRIEVAL TO EMPIRICALLY GROUNDED PROTOCOL EXECUTION. WITHIN EACH APPLICATION DIMENSION, SUB-SCORES ARE NORMALIZED TO $[0, 1]$ AND COMBINED BY THE LISTED WEIGHTS.

Level	Dimension	Sub-Metric	Method	Weight
Knowledge (I_K)	Static Retrieval	Accuracy	Deterministic	100%
Application (I_A)	Completeness	Step Integrity Param Detail	LLM-Judge	40%
			LLM-Judge	60%
	Correctness	Seq. Logic (co_order)	LLM-Judge	25%
		Method Consist. (co_method)	LLM-Judge	25%
		Param Consist. (co_param)	LLM-Judge	25%
		Chem. Safety (co_chem)	LLM-Judge	25%
	Effectiveness	Method Suit. (S_{method})	Signed Fit	25%
		Label Compat. (S_{label})	Rule-Based	25%
		Trans. Validity (S_{trans})	Physics Eq.	25%
		Time Efficiency (S_{time})	Physics Eq.	25%
Overall	Expert Score	Harmonic Mean	Aggregation	100%

are scored by an LLM judge (GPT-5.4) under a fixed rubric, since they require interpreting free-form text.

b) Effectiveness.: Effectiveness evaluates whether the extracted experimental choices satisfy TOC-specific feasibility constraints. Unlike Completeness and Correctness, it is scored by deterministic, non-LLM checks that read from the TOC feasibility database (Appendix A). From each generated protocol, we extract the clearing method, labeling target, fluorescent marker or labeling strategy, refractive-index condition, and total processing time. These fields are scored by four sub-metrics:

$$\text{Effectiveness} = \max\left(0, \frac{1}{4}(\hat{S}_{method} + \hat{S}_{label} + \hat{S}_{trans} + \hat{S}_{time})\right), \quad (2)$$

where $\hat{S}_{label}, \hat{S}_{trans}, \hat{S}_{time} \in [0, 1]$ are normalized by their maxima (6, 3, and 3), while the signed $\hat{S}_{method} = S_{method}/2.5 \in [-1, +1]$ can be negative when the chosen method conflicts with the scenario—so a poor method choice penalizes Effectiveness, which is clamped at 0.

Method suitability (S_{method} , **signed**, **max 2.5**). Each clearing method carries a *signed* six-dimensional capability vector: five axes (fluorescent-protein preservation, dye penetration, geometry, operational economy, safety) lie in $[-1, +1]$ —a positive entry means the method helps that objective, a negative entry that it actively harms it (an organic solvent quenches endogenous fluorescence, $c_{F,fp} < 0$)—while clearing power $c_{C,opt} \in [0, 1]$ is a non-negative magnitude. Each scenario is parsed by the teacher into a demand vector giving every axis a target $t_a \in [0, 1]$ and a constraint weight $w_a \in \{0, 0.5, 1\}$ ($w_a = 0$ when the scenario is silent), so the per-axis demand strength $d_a = w_a t_a$ makes unconstrained axes drop out. The two labeling axes are mutually exclusive: whichever of FP-preservation / dye-penetration has the larger weighted demand is emphasized ($\times 2$) and the other zeroed. Suitability is the need-weighted signed fit

$$S_{method} = 2.5 \cdot \frac{\sum_a d_a c_a}{\sum_a d_a} \in [-2.5, +2.5], \quad (3)$$

positive when the method matches the scenario’s binding constraints and negative when it conflicts (e.g. an FP-quenching

solvent for a reporter-line sample), replacing an earlier cosine variant that collapsed onto a single default method.

Label compatibility (S_{label} , **max 6**). This metric evaluates whether the proposed labeling strategy is compatible with the biological target and the selected clearing method:

$$S_{label} = \min(S_{target}, S_{marker}, S_{method-marker}). \quad (4)$$

S_{target} checks whether the marker matches the required labeling target; S_{marker} checks marker–fluorophore compatibility; and $S_{method-marker}$ checks fluorophore compatibility with the selected clearing method using the quenching matrix. Each term is scored on $[0, 6]$. The minimum operator implements a hard feasibility rule: failure of any one check makes the labeling strategy unusable.

Transparency validity (S_{trans} , **max 3**). This metric evaluates whether the refractive-index condition associated with the selected method is compatible with the tissue and sample scale. The score penalizes the gap between the tissue reference RI (RI_{tissue}) and the method RI (RI_{ref}) using a Gaussian tolerance σ_{RI} :

$$S_{trans} = 3 \cdot \exp\left(-\frac{(RI_{tissue} - RI_{ref})^2}{2\sigma_{RI}^2}\right). \quad (5)$$

S_{trans} is set to 0 when the selected method has no supported entry for the requested sample scale in the timing database.

Time efficiency (S_{time} , **max 3**). The proposed total processing time is compared with an evidence-supported window $[t_{min}, t_{max}]$. These windows are derived from TOC literature, 372 wet-lab experimental records, and expert review. Let Δt denote the deviation from this window, with $\Delta t = 0$ when the proposed time falls inside the interval. Out-of-window timing is penalized by

$$S_{time} = 3 \cdot \exp\left(-\frac{\Delta t^2}{2\tau^2}\right), \quad (6)$$

where τ is set deterministically to $0.2 \times$ the median clearing time.

All timing windows, RI references, and σ_{RI}/τ tolerances trace to literature, wet-lab records, and expert review (Appendix A). Effectiveness is the clamped mean of the four normalized sub-scores, scaled to $[0, 100]$ before entering the I_A bottleneck.

c) **Overall Expert Score.**: The final ClearEval Expert Score combines static knowledge and application-level feasibility using the harmonic mean:

$$\text{Score} = \frac{2I_K I_A}{I_K + I_A}. \quad (7)$$

This aggregation penalizes models that perform well on TOC knowledge questions but fail to generate experimentally feasible protocols.

IV. EXPERIMENTS

A. Models and Settings

We benchmark 13 LLMs across the GPT, Gemini, Claude, DeepSeek, GLM, and Qwen families. For all evaluated API

models we set temperature to 0.0 for both Knowledge and Application tasks to maximize reproducibility; reasoning variants additionally run in their native thinking mode. Exact provider snapshot strings, context windows, maximum output tokens, and API call dates are released in the repository configuration (Appendix C).

B. Main Results: The Knowledge-Execution Gap

Table II reveals a stark capability imbalance between static recall and wet-lab execution: Knowledge indices cluster at 79–91, yet the bottleneck Application Index I_A ranges from 56 down to 35. This is visualized in Fig. 2: Knowledge proficiency is uniformly high and tightly clustered, whereas Application performance is highly erratic. Models generally maintain high procedural Completeness (79 on average) and Correctness (69), but collapse on Effectiveness (50): for 12 of the 13 models Effectiveness is the binding constraint, the sole exception being Qwen3-14B, whose fluorescent-marker mismatches pull Correctness down to 35. (GLM-4.7-Think’s frequently truncated reasoning further depresses its Completeness to 51.) The dominant failure is label suitability (S_{label}): models propose fluorescent markers that miss the scenario’s required labeling targets (Sec. IV-D), exposing a severe knowledge–execution gap.

C. Calibration of the Automated Metrics

We validate the automated CCE scores against blinded expert consensus on 240 protocols (20 scenarios $\times$ 12 models). Because Pearson correlation alone ignores absolute bias, we report, per dimension, Pearson r , Spearman ρ , ICC(2,1), quadratic-weighted κ , and Krippendorff’s α (Table III; full 14-row sub-dimension breakdown in Appendix B). The aggregate CCE dimensions align well with experts (ICC 0.75/0.74/0.69 for Completeness/Correctness/Effectiveness; Overall $r = 0.79$). Agreement is *uneven* across the Effectiveness sub-metrics. Method suitability (S_{method})—the signed need-weighted fit, validated against expert method-suitability ratings we collected specifically—is among the best-calibrated (ICC 0.82, $r=0.83$), confirming that the signed axis space captures how experts judge method choice; timing (S_{time} , ICC 0.78) also tracks experts strongly and transparency (S_{trans} , 0.58) moderately. The exception is label compatibility (S_{label} , ICC 0.51): its hard-minimum over three binary checks is coarser than the graded expert scale, so although it is the dominant execution *failure* (Sec. IV-D) it is the most weakly-calibrated sub-metric. On the Correctness side all four checks calibrate well, including chemical safety (co_chem , ICC 0.80) once its rubric was revised to flag only introduced hazards—not the *absence* of unrequested safety caveats—and re-judged with GPT-5.4 (28% penalized).

D. Fine-Grained Diagnosis: Where Execution Fails

Averaging normalized sub-scores over all 253 scenarios and 13 models, the *reliably*-measured deficit concentrates in label suitability (Fig. 4). **Label suitability** (S_{label} , mean 0.32) is a hard minimum over a marker–target match and two

TABLE II

PERFORMANCE OF STATE-OF-THE-ART MODELS ON CLEAR-EVAL. EVALUATION COVERS KNOWLEDGE (MCQ) AND APPLICATION (OEQ). I_K REPRESENTS THE AVERAGE OF THE KNOWLEDGE SUBSET, I_A REPRESENTS THE MINIMUM SCORE ACROSS THE APPLICATION SUBSET, AND TOTAL IS THEIR HARMONIC MEAN. GLM-4.7-THINK’S THINKING-MODE REASONING FREQUENTLY EXHAUSTS THE OUTPUT-TOKEN BUDGET AND TRUNCATES ITS PROTOCOLS, LOWERING ITS COMPLETENESS. [†]GLM-4.7-THINK IS NOT COVERED BY THE EXPERT CALIBRATION (SEC. IV-C); [‡]GEMINI-3-FLASH IS SCORED ON 251/253 SCENARIOS (TWO UNPARSEABLE RESPONSES OMITTED, NOT A SYSTEMATIC FILTER).

Models	Knowledge (MCQ)				Application (OEQ)				Total
	Tissue	Reagent	Method	I_K	Com.	Cor.	Eff.	I_A	
<i>Proprietary Models</i>									
GPT-5.2-Fast	91.1	88.7	93.5	91.1	<u>97.0</u>	84.4	<u>55.2</u>	<u>55.2</u>	68.7
GPT-5.2-Think	<u>90.5</u>	<u>87.3</u>	<u>93.6</u>	<u>90.5</u>	<u>96.9</u>	85.4	54.4	54.4	<u>68.0</u>
Gemini-3-Flash [‡]	78.7	81.8	91.1	83.8	78.2	79.7	52.2	52.2	64.3
Gemini-3-Pro	88.8	77.2	86.7	84.2	83.8	<u>85.0</u>	46.6	46.6	60.0
Claude-4.6-Sonnet	87.6	82.9	94.9	88.5	98.6	77.4	56.2	56.2	68.7
<i>Open-Weight Models</i>									
GLM-4.7	79.3	80.2	93.5	84.3	77.5	73.2	53.4	53.4	65.4
GLM-4.7-Think [†]	78.7	70.5	90.7	80.0	51.1	67.2	47.6	47.6	59.7
DeepSeek-V3.2	79.9	77.9	91.7	83.1	78.6	68.3	41.6	41.6	55.4
DeepSeek-V3.2-Think	85.3	83.8	93.5	87.5	78.2	67.6	41.3	41.3	56.1
Qwen3-max	79.9	<u>87.3</u>	93.1	86.8	87.0	72.5	53.6	53.6	66.2
Qwen3-235B-A22B	77.5	81.2	89.1	82.6	76.8	52.4	50.1	50.1	62.3
Qwen3-32B	73.4	78.6	89.9	80.6	61.7	50.8	45.1	45.1	57.8
Qwen3-14B	73.4	77.8	88.4	79.9	58.4	35.3	46.8	35.3	48.9

TABLE III

AGREEMENT OF AUTOMATED VS. BLINDED EXPERT-CONSENSUS SCORES ON 240 PROTOCOLS (20 SCENARIOS $\times$ 12 MODELS). ICC=ICC(2,1); QWK=QUADRATIC-WEIGHTED κ ; α =INTERVAL KRIPPENDORFF. FULL 14-ROW BREAKDOWN IN APPENDIX B.

Dimension	Pearson r	Spearman	ICC	QWK	α
Completeness	0.77	0.79	0.75	0.70	0.75
Correctness	0.79	0.77	0.74	0.72	0.73
Effectiveness	0.71	0.72	0.69	0.68	0.69
Overall	0.79	0.79	0.74	0.73	0.74
<i>Effectiveness sub-metrics</i>					
S_{method} (signed fit)	0.83	0.77	0.82	0.77	0.82
S_{label} (rule)	0.51	0.48	0.51	0.51	0.51
S_{trans} (physics)	0.63	0.46	0.58	0.52	0.56
S_{time} (physics)	0.80	0.75	0.78	0.75	0.78

fluorophore-compatibility checks, and 53% of protocols score 0. Decomposing these zeros from the scorer’s own breakdown shows the binding cause is **marker–target mismatch**—the fluorescent markers models propose do not match the scenario’s required labeling targets. Every $S_{label}=0$ is a target mismatch; the quenching check often binds ($\sim 44\%$) but stays high (4.2/6) and never zeros. These zeros are *deterministic* labeling errors—directly verifiable, not scoring artifacts—so despite S_{label} ’s only moderate graded calibration (ICC 0.51, Sec. IV-C) this dominant deficit is a trustworthy execution failure. The re-judged *co_chem* (GPT-5.4) flags introduced hazards in $\sim 28\%$ of protocols. Transparency (S_{trans} , 0.64) and timing (S_{time} , 0.56) are moderately impaired, with small-sample timing failures: models over-estimate clearing time

(e.g. ~ 10 h for a $\leq 500 \mu\text{m}$ slice cleared in < 2 h) against the refined windows. S_{time} stays well-calibrated against experts (ICC 0.78, Sec. IV-C). Method suitability (S_{method} , 0.73) is comparatively high and now well-calibrated against experts (ICC 0.82, Sec. IV-C), so method *choice* is not the main deficit. In short, the most reliable constraint these LLMs miss is matching fluorescent markers to required targets.

Method path-dependence. Method *selection* is concentrated and largely scenario-independent (Fig. 3): iDISCO+ alone accounts for 43% of all 3,272 choices. Anchoring tracks scale—smaller models collapse onto a single default while stronger ones diversify (choice entropy 1.35 to 2.97 bits). The same familiarity prior drives labeling (canonical fluorophores over the required target, the S_{label} mismatch), a literature-frequency bias overriding scenario-specific reasoning.

E. Inference-Time Robustness Protocol

A natural follow-up is whether the knowledge–execution gap narrows once standard inference-time grounding is applied, rather than being an artifact of greedy decoding. We release a turnkey harness that scores grounding recipes against the greedy default on the full 253-scenario Application set, and instantiate it here on GPT-5.2-Fast—our strongest Application model—across three settings: the 1-shot default; +**KB-RAG**, which prepends a per-scenario knowledge-base context (standardized scenario slots, target→marker candidate sets, and qualitative method / compatibility records, *with numeric RI/time targets and gold protocols withheld so the answer key is never leaked*); and +**KB-RAG+self-check**, a deliberately cheap alternative to self-consistency in which the model

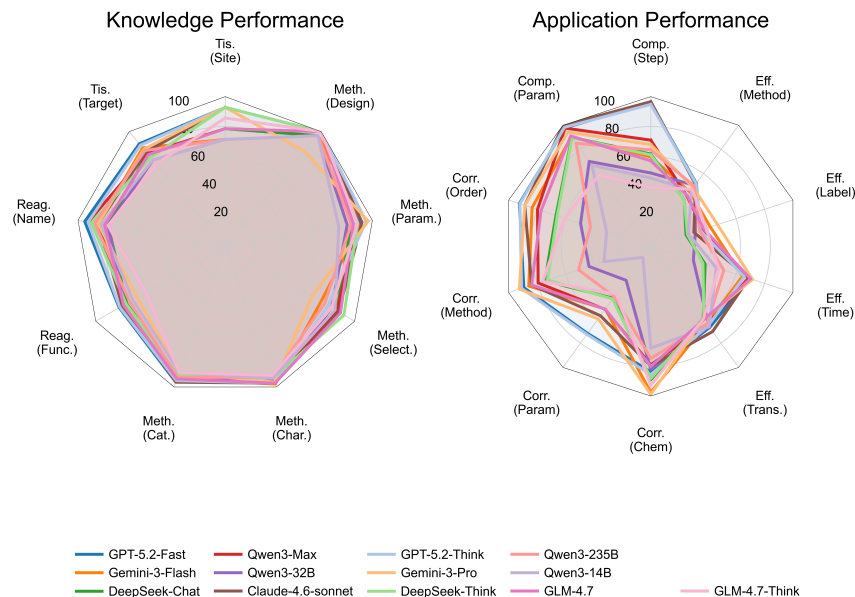


Fig. 2. Performance comparison of 13 LLMs on ClearEval. Left: Knowledge Performance (I_K) across domain attributes (Tissue, Reagent, Method). Right: Application Performance (I_A), across the CCE sub-metrics. MCQ profiles are tightly clustered; CCE scores are highly erratic.

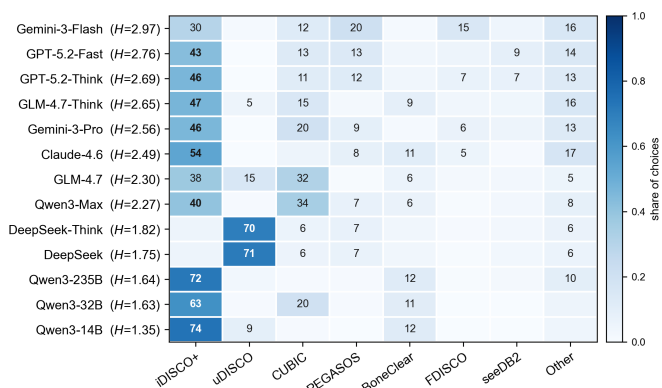


Fig. 3. Clearing-method selection per model (% of choices), rows sorted by choice entropy H . iDISCO+ dominates (43%); smaller models anchor hardest (Qwen3-14B 74%), DeepSeek on uDISCO—a training-frequency prior.

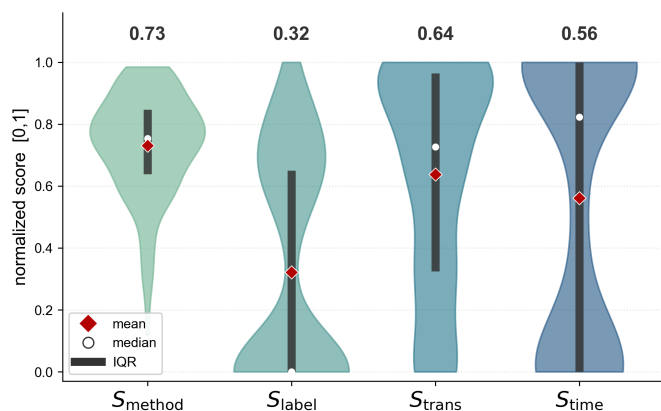


Fig. 4. Effectiveness sub-metric distributions over all 253×13 protocols, normalized to $[0, 1]$ (mean $\diamond$, median $\circ$, IQR bar; signed S_{method} rescaled so 0.5 is neutral). S_{label} (0.32) is the dominant failure, while S_{method} (0.73) is high and well-calibrated against experts (ICC 0.82).

runs one fixed feasibility checklist over its grounded answer and revises exactly once. The per-scenario RAG context is built deterministically from scenario metadata (identical across models) and released. Every setting is scored by the same calibrated CCE pipeline (Sec. IV-C), so the bottleneck I_A is directly comparable to Table II. We treat KB-RAG as an upper bound on grounding benefit and leave full literature-RAG and a paid multi-model sweep to an extended version.

Grounding sharply attacks the dominant failure mode: label suitability $\hat{S}_{label}$ nearly doubles ($0.32 \rightarrow 0.44 \rightarrow 0.59$), confirming that marker-target mismatch is an addressable knowledge gap rather than an intrinsic limit. Yet KB-RAG alone does not move the bottleneck (I_A : $55.2 \rightarrow 53.2$): to satisfy target-specific labeling the model is steered toward strong-clearing organic solvents, which incurs a compensating

loss on refractive-index and timing validity ($\hat{S}_{trans}$: $0.67 \rightarrow 0.57$; $\hat{S}_{time}$: $0.70 \rightarrow 0.53$) on soft tissue whose native RI favors aqueous media—a genuine trade-off that the balanced Effectiveness average exposes and that a single sub-metric would hide. The cheap self-check recovers most of this loss by re-verifying method-tier feasibility—it removes the RI/timing applicability zeros that KB-RAG had introduced—lifting I_A to 57.1 (+1.9 over the 1-shot default). The single-model probe thus shows that closing the marker-target gap is necessary but, under a feasibility-balanced metric, not sufficient to raise end-to-end protocol quality without an explicit feasibility check.

TABLE IV

INFERENCE-TIME GROUNDING ON GPT-5.2-FAST OVER THE 253 APPLICATION SCENARIOS. COM/COR/EFF/ I_A ARE PERCENTAGES (AS IN TABLE II); EFFECTIVENESS SUB-SCORES ARE NORMALIZED MEANS ($\hat{S}_{method} \in [-1, 1]$ SIGNED, OTHERS $\in [0, 1]$). KB-RAG WITHHOLDS ALL NUMERIC RI/TIME ANSWER KEYS; SELF-CHECK IS A SINGLE GROUNDED CHECKLIST REVISION, NOT SELF-CONSISTENCY.

Setting	Com	Cor	Eff	I_A	$\hat{S}_m$	$\hat{S}_{lab}$	$\hat{S}_{tr}$	$\hat{S}_{ti}$
1-shot (default)	97.0	84.4	55.2	55.2	0.52	0.32	0.67	0.70
+KB-RAG	93.8	78.8	53.2	53.2	0.58	0.44	0.57	0.53
+KB-RAG								
+self-check	92.1	79.7	57.1	57.1	0.59	0.59	0.60	0.51

V. DISCUSSION AND LIMITATIONS

Within TOC, ClearEval shows the bottleneck is not declarative retrieval but executable parameterization—most reliably, matching fluorescent markers to the required labeling targets, with refractive-index and timing errors secondary. Whether this pattern generalizes across wet-lab subdomains is an open question our single-subdomain design cannot settle.

External validity. ClearEval covers one subdomain (TOC), so our claims are about TOC protocol feasibility, not LLM scientific competence in general. Generalizing requires recurating subdomain-specific knowledge bases (refractive indices, incubation windows, compatibility matrices), which is non-trivial (Appendix A); we offer CCE as a transferable recipe, not a finished multi-domain benchmark.

Complementarity. Depth-first ClearEval and breadth-first benchmarks such as BioProBench [12] are complementary: the former contributes parameter-level feasibility scoring within a subdomain, the latter broad coverage—together a fuller picture than either alone.

Toward grounded agents. For LLMs to become viable in specialized wet-lab domains, they must internalize experimental constraints within their reasoning loops; ClearEval can serve as a pre-experimental feasibility gate before irreversible reagent commitment.

Limitations. ClearEval remains a static paradigm: real-world wet labs involve stochastic variables (reagent degradation, tissue heterogeneity) hard to model in silico, and our expert grounding is literature-bounded rather than exhaustively re-measured. Future work should integrate dynamic hardware feedback so agents iteratively optimize protocols from real-time signals.

VI. CONCLUSION

We presented ClearEval, an expert-curated benchmark for TOC protocol design that shifts evaluation from semantic similarity to empirically grounded functional validation. By leveraging a dual-track pipeline and a data-driven CCE framework, ClearEval exposes a significant knowledge-execution gap in current LLMs on TOC protocol design. Models excel at knowledge recall but struggle with the marker–target matching and physical-feasibility constraints required for wet-lab execution. ClearEval and the transferable CCE recipe aim to

facilitate reliable scientific agents, starting with tissue-clearing workflows.

APPENDIX A

KNOWLEDGE-BASE CURATION AND EXPERT GROUNDING

Expert panel and consensus. The knowledge bases and CCE rubric were curated by a panel of tissue-clearing wet-lab experts with a knowledge-graph specialist. Application protocols were scored blind against the rubric then reconciled by consensus; released scores are per-protocol consensus values.

Knowledge bases. Grounding is encoded in seven JSON knowledge bases: reference refractive index for 18 clearing methods (CUBIC, ScaleS, iDISCO+, FDISCO, SeeDB2, PEGASOS, SWITCH, uDISCO, ...); per-method RI tolerance σ_{RI} ; per-method incubation-time tolerance τ ; a timing base of 84 (method, sample-tier) rows giving the viable window $[t_{min}, t_{max}]$; a marker $\times$ method fluorophore-compatibility matrix; a tissue table; and tissue-RI references.

Provenance. Each timing row records a literature source key, source URL, evidence locator, confidence level, and review status, giving an audit trail from each parameter to its literature basis. RI references cite their originating protocol (e.g., CUBIC RI=1.46, Susaki et al., *Nat. Protoc.*). Timing tolerances are derived deterministically as $\tau=0.2\times$ the median clearing time. Parameters are literature-bounded and refined by expert review rather than exhaustively re-measured.

Operationalization. The Effectiveness terms read directly from these bases: S_{method} from the signed method-capability and scenario-demand vectors (a signed need-weighted fit); S_{label} from the fluorophore-compatibility matrix (hard zero on quenching); S_{trans} from RI references and σ_{RI} , gated to zero when a (method, tier) pair is unsupported; and S_{time} from $[t_{min}, t_{max}]$ and τ .

APPENDIX B

JUDGE CALIBRATION DETAILS

Table V reports the full per-sub-dimension agreement between automated and blinded expert-consensus scores ($n=240$), including the now well-calibrated S_{method} . The two LLM-judged dimensions use a fixed rubric (eval_oeq_teacher_rubric.txt, eval_completeness_checklist.txt, released in the repository) at temperature 0.0. Because raw per-annotator sheets were not preserved, we report expert-consensus-vs-automated agreement rather than human–human inter-annotator reliability.

APPENDIX C

REPRODUCIBILITY AND ARTIFACT MANIFEST

Artifacts and determinism. The repository releases the 253 Application scenarios and 2,588 MCQs, the seven knowledge bases, all 13 model generations and their evaluation results, the generation/judge prompts, and the scoring/plotting code (English README, requirements.txt). All evaluated API models run at temperature 0.0; since hosted models are non-deterministic and snapshot aliases drift, we release the

TABLE V
FULL PER-DIMENSION AGREEMENT OF AUTOMATED VS.
EXPERT-CONSENSUS SCORES ($n=240$). ICC=ICC(2,1);
QWK=QUADRATIC-WEIGHTED κ ; α =INTERVAL KRIPPENDORFF.

Dimension	r	ρ	ICC	QWK	α
c_step	0.71	0.72	0.70	0.54	0.70
c_param	0.71	0.71	0.68	0.60	0.67
Completeness	0.77	0.79	0.75	0.70	0.75
co_order	0.66	0.65	0.61	0.58	0.60
co_method	0.83	0.81	0.79	0.77	0.79
co_param	0.55	0.56	0.53	0.48	0.53
co_chem	0.83	0.69	0.80	0.66	0.79
Correctness	0.79	0.77	0.74	0.72	0.73
S_{method}	0.83	0.77	0.82	0.77	0.82
S_{label}	0.51	0.48	0.51	0.51	0.51
S_{trans}	0.63	0.46	0.58	0.52	0.56
S_{time}	0.80	0.75	0.78	0.75	0.78
Effectiveness	0.71	0.72	0.69	0.68	0.69
Overall	0.79	0.79	0.74	0.73	0.74

raw responses—with provider snapshots and API-call dates—so results are replayable even when regeneration is not bit-identical.

Inference-time protocol (Sec. IV-E). KB-RAG prepends per-scenario cards—scenario slots, target→marker candidate sets, and qualitative method/compatibility records—built deterministically from scenario metadata, with numeric RI/time targets and gold protocols withheld; self-check runs one fixed feasibility checklist over the grounded answer and revises exactly once (a cheap alternative to $k=5$ self-consistency). The released harness also exposes few-shot and self-consistency recipes; the single-model GPT-5.2-Fast instantiation is reported in Table IV.

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